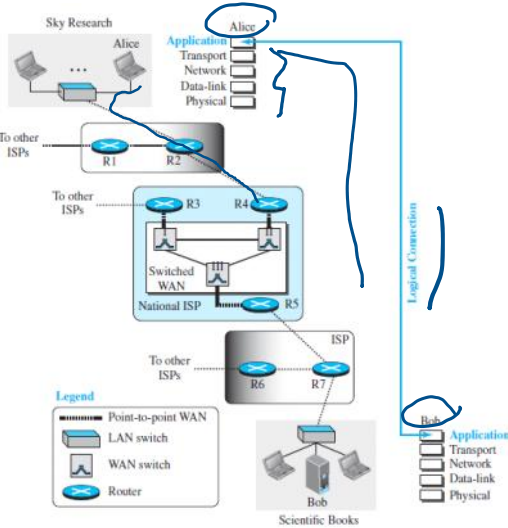
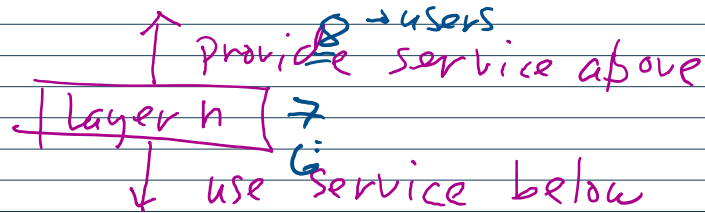


Application Layer (L7)



=> Services are user-centric

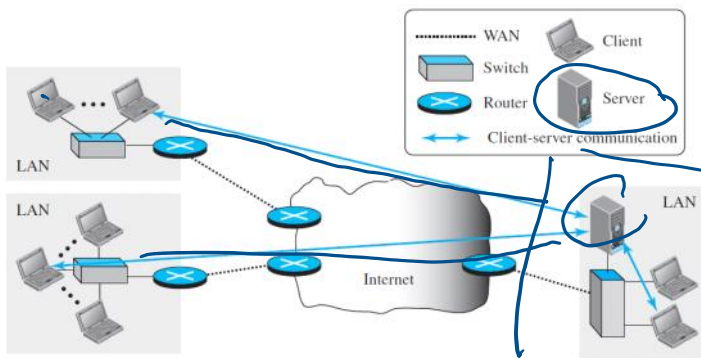


=> protocols may be

- (a) standard: HTTP, FTP, Telnet, DNS
- (b) non-standard: games, multimedia,

Client-Server Paradigm

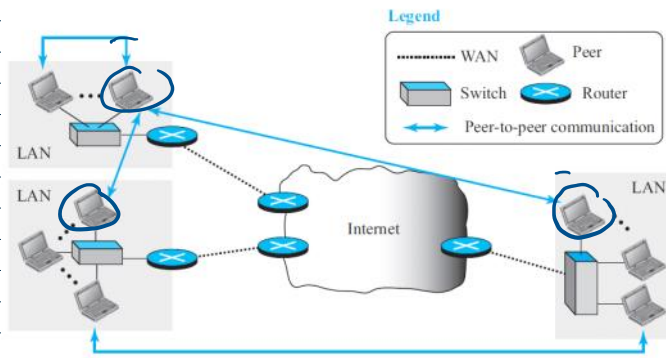
=> bottleneck on server — DOS
=> needs a "Provider" to host a server



Peer-to-Peer Paradigm (P2P)

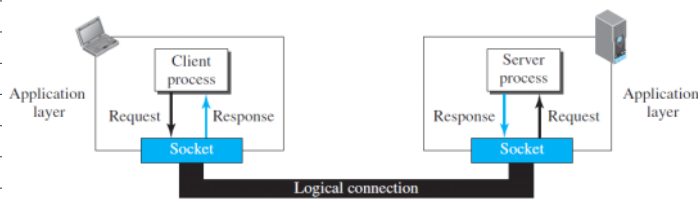
=> server is for coordination — BitTorrent
=> issues: security, not all services run

⇒ server is for coordination - BitTorrent
 ⇒ issues: security, not all services can be made P2P



Socket Programming

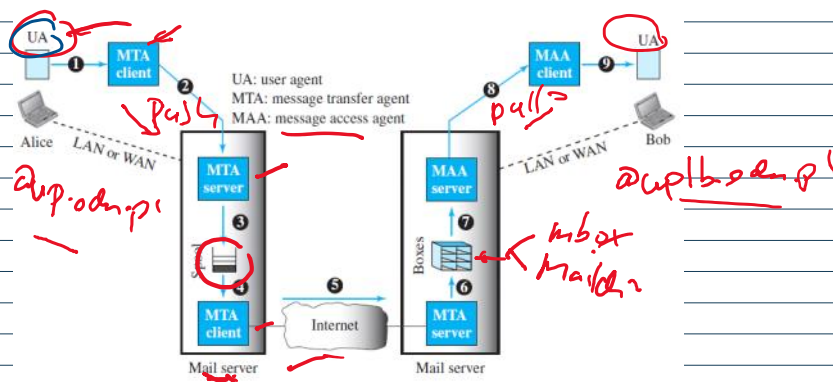
↳ "socket" abstraction socket addr



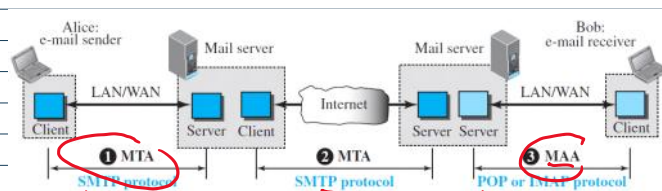
IPaddr:port

Electronic Mail (E-mail)

⇒ no immediate response is required



Protocols for e-mail



Simple Mail Transfer Protocol (SMTP)

Commands

Keyword	Argument(s)	Description
HELO	Sender's host name	Identifies itself
MAIL FROM	Sender of the message	Identifies the sender of the message
RCPT TO	Intended recipient	Identifies the recipient of the message
DATA	Body of the mail	Sends the actual message
QUIT		Terminates the message
RSET		Aborts the current mail transaction
VERFY	Name of recipient	Verifies the address of the recipient
NOOP		Checks the status of the recipient
TURN		Switches the sender and the recipient
EXPN	Mailing list	Asks the recipient to expand the mailing list
HELP	Command name	Asks the recipient to send information about the command sent as the argument
SEND FROM	Intended recipient	Specifies that the mail be delivered only to the terminal of the recipient, and not to the mailbox
SMOL FROM	Intended recipient	Specifies that the mail be delivered to the terminal <i>or</i> the mailbox of the recipient
SMAL FROM	Intended recipient	Specifies that the mail be delivered to the terminal <i>and</i> the mailbox of the recipient

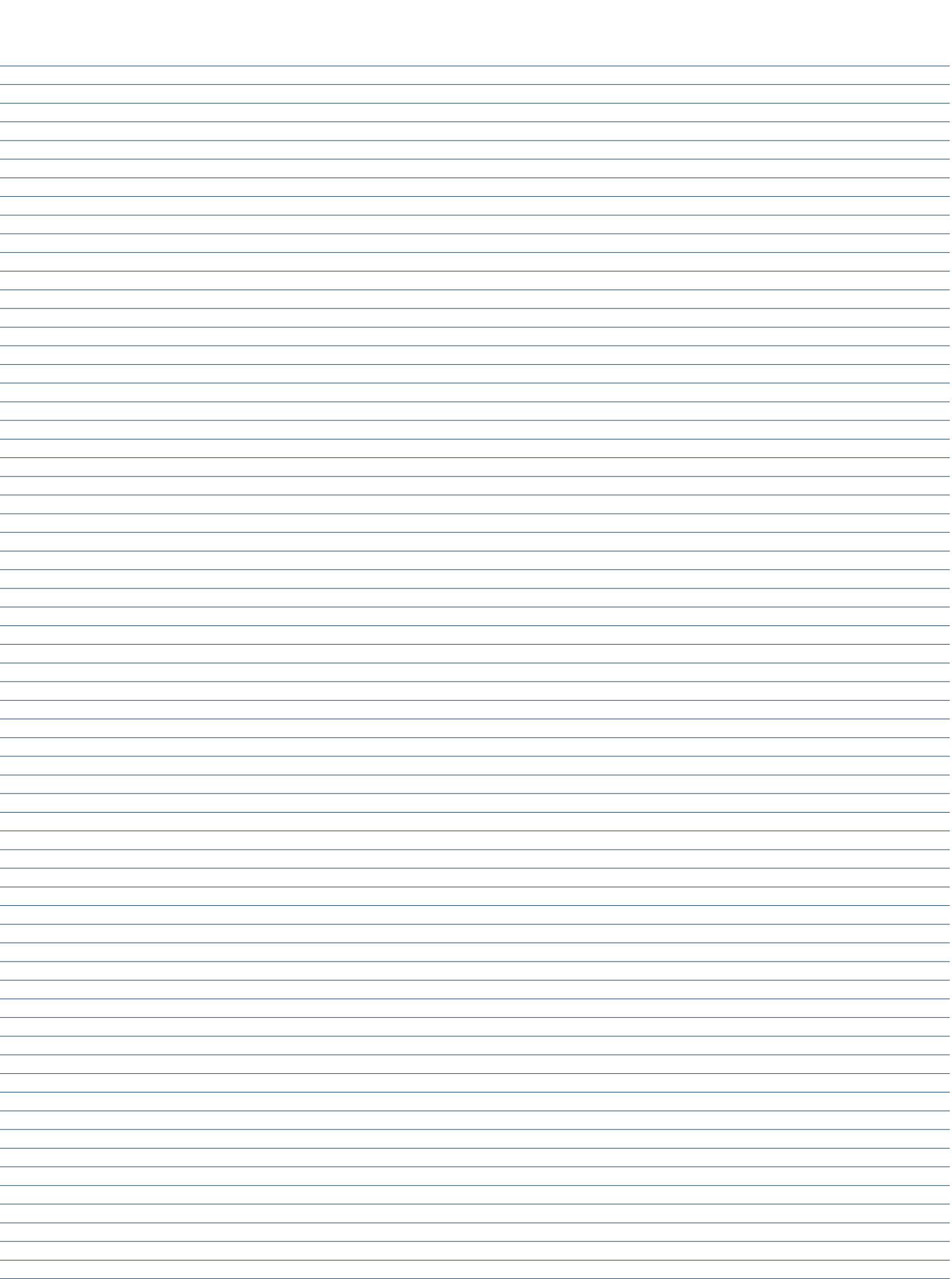
Responses

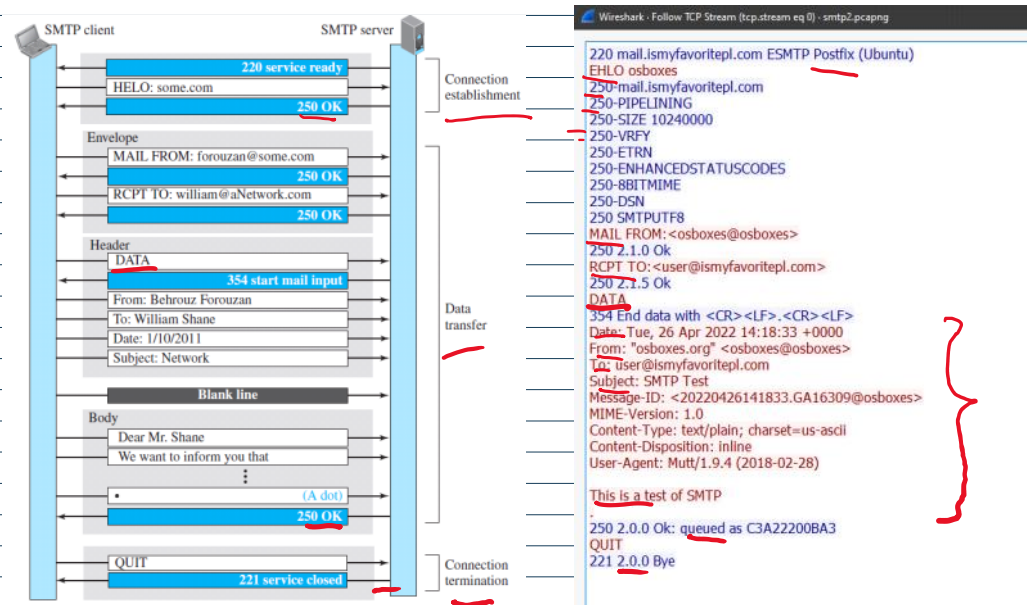
Code	Description
Positive Completion Reply	
211	System status or help reply
214	Help message
220	Service ready
221	Service closing transmission channel
250	Request command completed
251	User not local; the message will be forwarded
Positive Intermediate Reply	
354	Start mail input
Transient Negative Completion Reply	
421	Service not available
450	Mailbox not available
451	Command aborted; local error
452	Command aborted; insufficient storage
Permanent Negative Completion Reply	
500	Syntax error; unrecognized command

200
421

Code	Description
501	Syntax error in parameters or arguments
502	Command not implemented
503	Bad sequence of commands
504	Command temporarily not implemented
550	Command is not executed; mailbox unavailable
551	User not local
552	Requested action aborted; exceeded storage location
553	Requested action not taken; mailbox name not allowed
554	Transaction failed

Phases



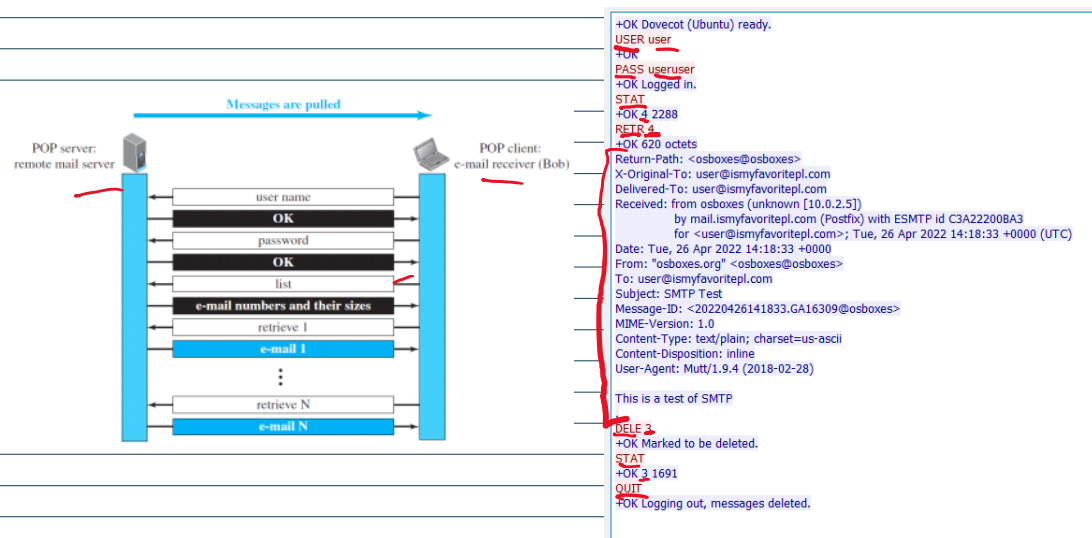


(Aside: Sample Server and Client) .2.4 .2.5

```
osboxes@osboxes:~$ sudo netstat -tnlp
Active Internet connections (only servers)
Proto Recv-Q Send-Q Local Address   Foreign Address State      PID/Program name
tcp        0      0 0.0.0.0:22       0.0.0.0:*      LISTEN    2944/sshd
tcp        0      0 0.0.0.0:23       0.0.0.0:*      LISTEN    1033/inetd
tcp        0      0 0.0.0.0:25       0.0.0.0:*      LISTEN    2944/named
tcp        0      0 0.0.0.0:110      0.0.0.0:*      LISTEN    23178/dovecot
tcp        0      0 0.0.0.0:143      0.0.0.0:*      LISTEN    23178/dovecot
tcp6       0      0 :::22           :::*           LISTEN    1443/sshd
tcp6       0      0 :::23           :::*           LISTEN    1033/inetd
tcp6       0      0 :::25           :::*           LISTEN    2944/named
tcp6       0      0 :::110          :::*           LISTEN    23178/dovecot
tcp6       0      0 :::143          :::*           LISTEN    23178/dovecot
osboxes@osboxes:~$
```

```
osboxes@osboxes:~$ cat ~/.muttrc
set mbox_type = Maildir
set folder = ~/Maildir
set smtp_url = 'smtp://mail.ismyfavoritepl.com:25'
set pop_host = 'pop://user:useruser@mail.ismyfavoritepl.com:110'
osboxes@osboxes:~$
```

Post Office Protocol (POP3)



Internet Mail Access Protocol (IMAP4)


```

a4 OK List completed (0.001 + 0.000 secs).
a3 EXAMINE INBOX
* FLAGS (Unanswered \Flagged \Deleted \Seen \Draft)
* OK [PERMANENTFLAGS {}] Read-only mailbox.
* 3 EXISTS
* 0 RECENT
* OK [UIDVALIDITY 1650981658] UIDs valid
* OK [UIDNEXT 5] Predicted next UID
a3 OK [READ-ONLY] Examine completed (0.001 + 0.000 secs).
a4 FETCH 3 BODY[] (620)
Return-Path: <osboxes@osboxes>
X-Original-To: user@ismyfavoritepl.com
Delivered-To: user@ismyfavoritepl.com
Received: from osboxes (unknown [10.0.2.5])
  by mail.ismyfavoritepl.com (Postfix) with ESMTP id C3A22200BA3
  for <user@ismyfavoritepl.com>; Tue, 26 Apr 2022 14:18:33 +0000 (UTC)
From: "osboxes.org" <osboxes@osboxes>
To: user@ismyfavoritepl.com
Subject: SMTP Test
Message-ID: <20220426141833.GA16309@osboxes>
MIME-Version: 1.0
Content-Type: text/plain; charset=us-ascii
Content-Disposition: inline
User-Agent: Mut/1.9.4 (2018-02-28)

This is a test of SMTP

a4 OK Fetch completed (0.001 + 0.000 secs).
LOGOUT
LOGOUT BAD Error in IMAP command : Unknown command (1716095.920 + 0.000 secs).
a5 LOGOUT
* BYE Logging out
a5 OK Logout completed (0.001 + 0.000 secs).
Connection closed by foreign host.
osboxes@osboxes:~$ _

```

```

Winchark - Follow TCP Stream (tcp.stream eq 0) - imap.pcapng
* OK [CAPABILITY IMAP4rev1 LITERAL+ SASL-IR LOGIN-REFERRALS ID ENABLE IDLE STARTTLS AUTH=PLAIN] Dovecot (Ubuntu) ready.
a1 LOG IN user user@ismyfavoritepl.com
* OK [CAPABILITY IMAP4rev1 LITERAL+ SASL-IR LOGIN-REFERRALS ID ENABLE IDLE SORT SORT=DISPLAY THREAD=REFERENCES THREAD=REFS
  SEARCH=ORDEREDSUBJECT MULTIAPPEND URL-PARTIAL CATENATE UNSELECT CHILDREN NAMESPACE UIDPLUS LIST-EXTENDED I18NLEVEL=1 CONDSTORE QRESYNC
  SEARCH ESORT SEARCHES WITHIN CONTEXT=SEARCH LIST-STATUS BINARY MOVE] Logged in
a2 LIST "" (HasNoChildren) "*" INBOX
* LIST (HasNoChildren) "*" INBOX
a2 OK List completed (0.001 + 0.000 secs).
a3 EXAMINE INBOX
* FLAGS (Unanswered \Flagged \Deleted \Seen \Draft)
* OK [PERMANENTFLAGS {}] Read-only mailbox.
* 3 EXISTS
* 0 RECENT
* OK [UIDVALIDITY 1650981658] UIDs valid
* OK [UIDNEXT 5] Predicted next UID
a3 OK [READ-ONLY] Examine completed (0.001 + 0.000 secs).
a4 FETCH 3 BODY[] (620)
Return-Path: <osboxes@osboxes>
X-Original-To: user@ismyfavoritepl.com
Delivered-To: user@ismyfavoritepl.com
Received: from osboxes (unknown [10.0.2.5])
  by mail.ismyfavoritepl.com (Postfix) with ESMTP id C3A22200BA3
  for <user@ismyfavoritepl.com>; Tue, 26 Apr 2022 14:18:33 +0000 (UTC)
From: "osboxes.org" <osboxes@osboxes>
To: user@ismyfavoritepl.com
Subject: SMTP Test
Message-ID: <20220426141833.GA16309@osboxes>
MIME-Version: 1.0
Content-Type: text/plain; charset=us-ascii
Content-Disposition: inline
User-Agent: Mut/1.9.4 (2018-02-28)

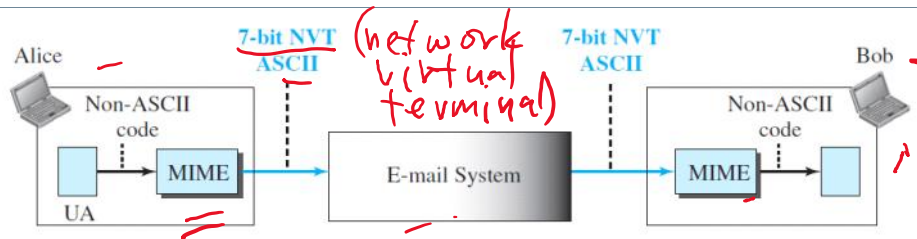
This is a test of SMTP

a4 OK Fetch completed (0.001 + 0.000 secs).
LOGOUT
LOGOUT BAD Error in IMAP command : Unknown command (1716095.920 + 0.000 secs).
a5 LOGOUT
* BYE Logging out
a5 OK Logout completed (0.001 + 0.000 secs).

```

Mult purpose Internet Mail Extensions (MIME)

=> email can send only 7-bit ASCII !!
 How can we send/binary files??
 attach



MIME Headers

MIME headers	E-mail header	
	E-mail body	
	MIME-Version: 1.1 <u>Content-Type</u> : type/subtype Content-Transfer-Encoding: encoding type Content-ID: message ID Content-Description: textual explanation of nontextual contents	

Content Type

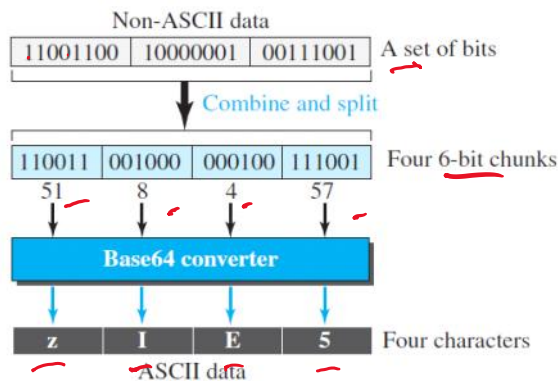
Type	Subtype	Description
<u>Text</u>	Plain	Unformatted
	HTML	HTML format (see Appendix C)
	Mixed	Body contains ordered parts of different data types
Multipart	Parallel	Same as above, but no order
	Digest	Similar to Mixed, but the default is message/RFC822
	Alternative	Parts are different versions of the same message
Message	RFC822	Body is an encapsulated message
	Partial	Body is a fragment of a bigger message
	External-Body	Body is a reference to another message
Image	JPEG	Image is in JPEG format
	GIF	Image is in GIF format
Video	MPEG	Video is in MPEG format
Audio	Basic	Single channel encoding of voice at 8 KHz
	PostScript	Adobe PostScript

Type	Subtype	Description
Text	Plain	Unformatted
	HTML	HTML format (see Appendix C)
Multipart	Mixed	Body contains ordered parts of different data types
	Parallel	Same as above, but no order
	Digest	Similar to Mixed, but the default is message/RFC822
	Alternative	Parts are different versions of the same message
Message	RFC822	Body is an encapsulated message
	Partial	Body is a fragment of a bigger message
	External-Body	Body is a reference to another message
Image	JPEG	Image is in JPEG format
	GIF	Image is in GIF format
Video	MPEG	Video is in MPEG format
Audio	Basic	Single channel encoding of voice at 8 KHz
	PostScript	Adobe PostScript
	Octet-stream	General binary data (eight-bit bytes)

encoding

Type	Description
7-bit	NVT ASCII characters with each line less than 1000 characters
8-bit	Non-ASCII characters with each line less than 1000 characters
Binary	Non-ASCII characters with unlimited-length lines
Base64	6-bit blocks of data encoded into 8-bit ASCII characters
Quoted-printable	Non-ASCII characters encoded as an equal sign plus an ASCII code

Base 64



Sample email

```

MIME-Version: 1.0
From: Joseph Anthony Hermocilla <jachermocilla@gmail.com>
Date: Wed, 27 Apr 2012 11:00:26 +0800
Message-ID: <CA3BNfajgkRul4Hvuse-j5NfTfOn5AuC0DteF0PPm+Xw4z2eCb@gmail.gmail.com>
Subject: Test Email With Non-ASCII Attachment
To: Joseph Anthony Hermocilla <jachermocilla@up.edu.ph>
Content-Type: multipart/mixed; boundary="00000000000007c370205dd9a1a2b"

--00000000000007c370205dd9a1a2b
Content-Type: multipart/alternative; boundary="00000000000007c370205dd9a1a2b"

--00000000000007c370205dd9a1a2b
Content-Type: text/plain; charset="UTF-8"

Boom!

--
*Joseph Anthony C. Hermocilla*
*PhD in Computer Science Student*
Institute of Computer Science
College of Arts and Sciences
University of the Philippines Los Banos
https://jachermocilla.org

--00000000000007c370205dd9a1a2b
Content-Type: text/html; charset="UTF-8"

<div dir="ltr">Boom!<br clear="all"><div><br></div>--<br><div dir="ltr" class="gmail_signature" data-smartmail="gmail_signature"><div dir="ltr"><div><div dir="ltr"><div><div
dir="ltr"><div><b>Joseph Anthony C. Hermocilla</b>
<br><i>PhD in Computer Science Student</i> <br>Institute of Computer Science
<br>College of Arts and Sciences
<br>University of the Philippines Los Banos
<br>a href="https://jachermocilla.org/" target="_blank" https://jachermocilla.org/a/
</div></div></div></div></div></div></div>

--00000000000007c370205dd9a1a2b--
--00000000000007c370205dd9a1a2b
Content-Type: image/png; name="avatar2.png"
Content-Disposition: attachment; filename="avatar2.png"
Content-Transfer-Encoding: base64
Content-ID: <f_12g1w7p00
X-Attachment-Id: f_12g1w7p00

--00000000000007c370205dd9a1a2b--

```

What about Web-based Email?

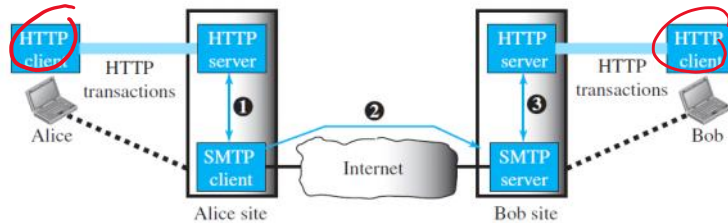
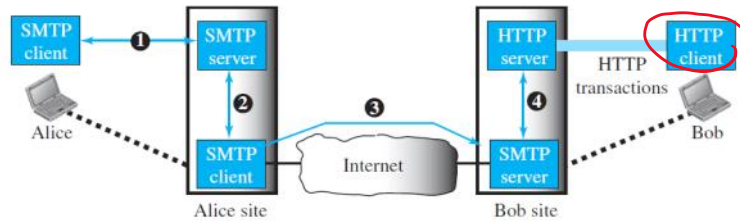
liham.wplb.edu.ph

At home

What about Uplb.edu Email:

liham.uplb.edu.ph

Other MUA



- Outlook
- Thunderbird
- MUA
- Pine
- Evolution

Email and DNS

```
jachermocilla@atlas:~$ dig up.edu.ph MX
; <<>> DiG 9.11.3-ubuntu1.17-Ubuntu <<>> up.edu.ph MX
;; global options: +cmd
;; Got answer:
;; ->>HEADER<<- opcode: QUERY, status: NOERROR, id: 43785
;; flags: qr rd ra; QUERY: 1, ANSWER: 7, AUTHORITY: 0, ADDITIONAL: 1
;; OPT PSEUDOSECTION:
;; EDNS: version: 0, flags:;, udp: 512
;; QUESTION SECTION:
;; up.edu.ph.
;; IN MX
;; ANSWER SECTION:
up.edu.ph. 235 IN MX 10 aspmx3.googlemail.com.
up.edu.ph. 235 IN MX 1 aspmx.l.google.com.
up.edu.ph. 235 IN MX 5 alt2.aspmx.l.google.com.
up.edu.ph. 235 IN MX 5 alt1.aspmx.l.google.com.
up.edu.ph. 235 IN MX 10 aspmx2.googlemail.com.
up.edu.ph. 235 IN MX 10 aspmx4.googlemail.com.
up.edu.ph. 235 IN MX 10 aspmx5.googlemail.com.
;; Query time: 21 msec
;; SERVER: 8.8.8.8#53(8.8.8.8)
;; WHEN: Thu Apr 28 08:09:59 PST 2022
;; MSG SIZE rcvd: 217

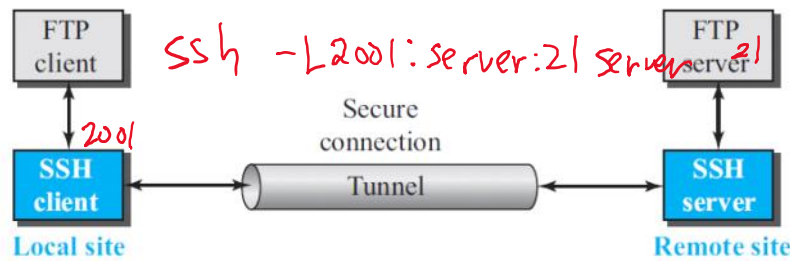
jachermocilla@atlas:~$ nc aspmx3.googlemail.com 25
220 mx.google.com ESMTP w4-20020a9d77c4000000b006054a0eabfdsi1822836otl.113 - gsmt
```

```
2023-05-05 08:06:23 ● ubuntu2204 in ~
➜ dig uplb.edu.ph MX
; <<>> DiG 9.18.12-0ubuntu0.22.04.1-Ubuntu <<>> uplb.edu.ph MX
;; global options: +cmd
;; Got answer:
;; ->>HEADER<<- opcode: QUERY, status: NOERROR, id: 48767
;; flags: qr rd ra; QUERY: 1, ANSWER: 1, AUTHORITY: 0, ADDITIONAL: 1
;; OPT PSEUDOSECTION:
;; EDNS: version: 0, flags:;, udp: 65494
;; QUESTION SECTION:
;; uplb.edu.ph.
;; IN MX
;; ANSWER SECTION:
uplb.edu.ph. 131 IN MX 0 new-rimas.uplb.edu.
;; Query time: 0 msec
;; SERVER: 127.0.0.53#53(127.0.0.53) (UDP)
;; WHEN: Fri May 05 08:06:28 PST 2023
;; MSG SIZE rcvd: 66

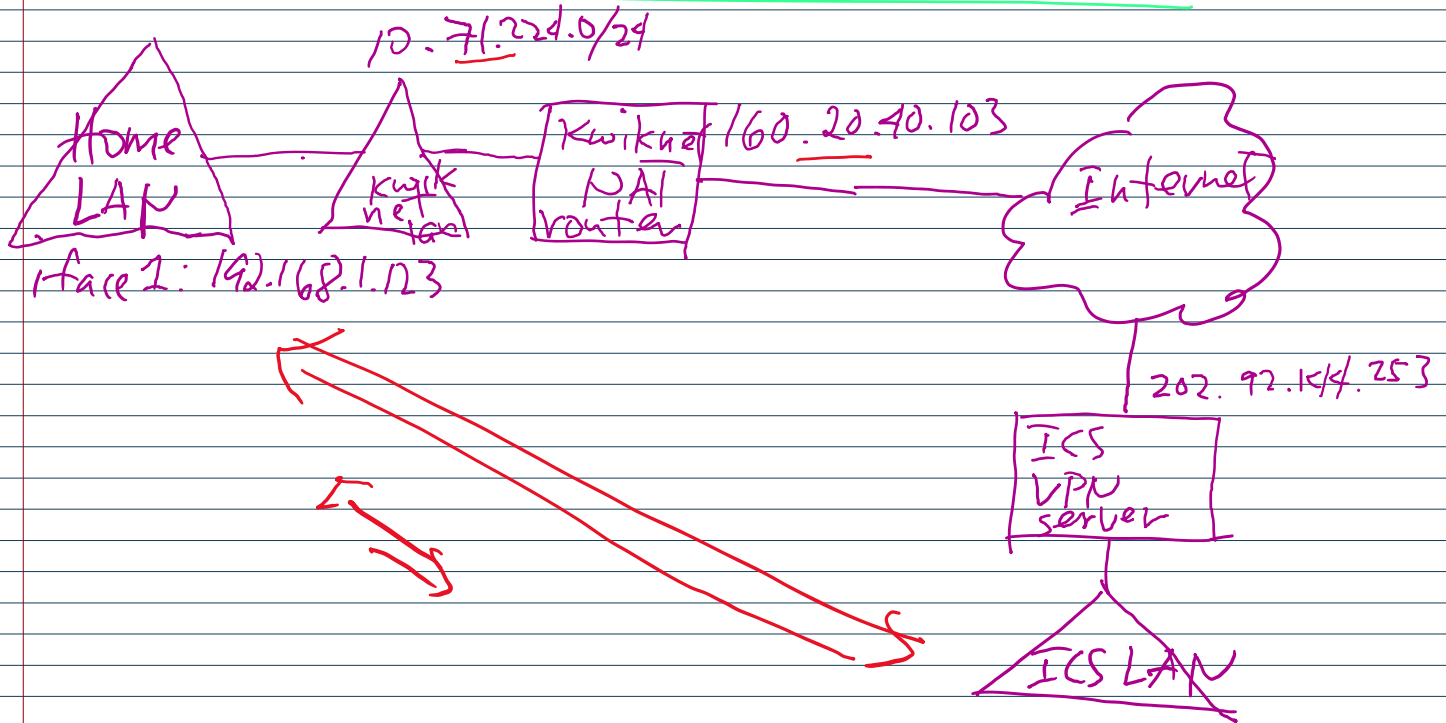
2023-05-05 08:06:28 ● ubuntu2204 in ~
➜ nc new-rimas.uplb.edu.ph 25
220 new-rimas.uplb.edu.ph ESMTP Postfix (Ubuntu)
```

Telnet / SSH

ph.



Virtual Private Network (VPN)



Remote Desktop : RDP, VNC, others

<https://blog.devolutions.net/2019/01/updated-2019-most-popular-free-remote-desktop-solution/>

	AnyDesk	Chrome RDP	Windows RDP	Ultra VNC	Remote Utilities	Team Viewer	Connect Wise Control	Lite Manager	Zoho Assist
Drag and Drop	*	✓	✓	✓	✓	✓	*	✓	*
File Transfer	✓	✓	***	✓	✓	✓	*	✓	*
Copy/Paste	✓	✓	✓	✓	✓	✓	*	✓	✓
Session Recording	✓	✓	✓	✓	✓	✓	✓	*	*
Screen Sharing (1)	✓	✓	✓	✓	✓	✓	✓	✓	*
Share Control	✓	✓	✓	✓	✓	✓	✓	✓	✓
Audio Support (2)	✓	✓	✓	✓	✓	✓	*	✓	✓
Audio VoIP	✓	✓	✓*****	✓	✓	✓	*	*	✓
Unattended Access (3)	✓	✓	✓	✓	✓	✓	*	✓	*
Supports Mobile App	✓	✓	✓	✓	✓	✓	✓	✓	*
Portable Version (4)	✓	✓	✓	✓	✓	✓	✓	✓	✓
Chat Across Computer	✓	✓	✓	✓	✓	✓	*	✓	✓
Clipboard Synchronization (5)	✓	✓	✓	✓	✓	✓	*	✓	✓
Multi-Monitor Support	✓	✓	✓	✓	✓	✓	*	✓	*

Lined paper template with horizontal ruling lines.